

SULIT



SECP2523 DATABASE

SESSION 2024/2025 - SEMESTER 1

ALTERNATIVE ASSESSMENT REPORT: PHASE 1

Name	TAN ZHI MING
IC No. / Matric No.	A23CS0189
Year / Program	2/SECPH
Section	2
Lecturer Name	Dr Seah Choon Sen
Case Study	Administrative
System Name	Kada-eServe
Group Name	TechAway

SECTION A (SYSTEM'S OVERVIEW)		
1.1	Overall Description	
		<< <i>Describe the system's overview.</i> >>
1.2	Project Weaknesses or Improvement	
		<< <i>List the project's weaknesses or improvement that can be made to the proposed system. Provide an explanation of the proposed module to overcome the weaknesses.</i> >>
1.3	Proposed Module	
		<< <i>Describe a new module to overcome the project's weaknesses.</i> >>
SECTION B (DATABASE PLANNING AND DESIGN)		
2.1	Conceptual ERD	
		<< <i>Draw a conceptual ERD to represent the proposed module.</i> >>
2.2	Logical ERD	
		<< <i>Provide logical ERD of your module in the proposed system.</i> >>
2.3	Data Dictionary for Logical ERD	
		<< <i>Provide the data dictionary based on the logical ERD in Subsection 2.2.</i> >>
2.4	Relational Database Schema	
		<< <i>Provide all relations database schema of your proposed module.</i> >>
2.5	Normalization	
		<< <i>Illustrate the normalization process up until BCNF to every relation listed in 2.5. At the end of this section, list all normalized relations achieved.</i> >>

REPORT: PHASE 1

SECTION A – SYSTEM’S OVERVIEW

1.1 Overall Description

Group Members’ Name	Matric No	Module
TAN ZHI MING	A23CS0189	Administrative
SABRINA HENG WEI QI	A23CS0265	Member Management
POH LOK YEE	A23CS0262	Reporting
GUI KAH SIN	A23CS0080	Loan Management
BRENDAN CHIA YAN FEI	A23CS0021	Savings and Dividend Management

The proposed KADA system aims to address the inefficiencies of the existing system, particularly in managing membership registrations, loan applications and financial statements, which currently rely on manual processing. Problems include time-consuming hard copy registration forms, delays in loan processing and manual generation of financial statements, all of which add to errors and workload. In addition, current websites lack basic features to support these processes. The KADA system will introduce a digital platform to simplify registration, automate loan tracking and financial statement distribution, and enhance website functionality to improve efficiency and user experience.

1.2 Project Weakness and Improvement

The management module of the current KADA system has several shortcomings. First, the traditional paper-based approach requires members to contact the relevant department of KADA to view their transfer records, which is inconvenient and time-consuming. Secondly, the system cannot automatically generate statistical reports, need manual compilation and analysis, both laborious and error prone. After that, members are unable to track the status of their applications in real time, resulting in a lack of transparency and frequent inquiries. Finally, a significant disadvantage of the paper-based approach in the current management module is the inefficiency and risk of errors in validating applications. Administrators must manually check application details against a variety of criteria, such as eligibility, completeness, and accuracy. This process is time-consuming, prone to human error, and can cause processing delays. In addition, paper-based approaches lack a centralized way to track which applications have been validated, leading to potential mismanagement or duplication of effort. These issues highlight the need for a more automated and reliable verification process. To improve the module, the introduction of a digital interface with direct access to transmission records, automatic generation of statistical reports, and real-time application status updates will greatly improve efficiency and user satisfaction.

1.3 Proposed Module

To solve the limitations of the current management module, four key modules are proposed: transaction module, transaction history module, system report module and validation module. Together, these modules will improve the efficiency and transparency of the KADA system.

Transaction Module

The transaction module is designed to allow administrators to enter and manage transaction records digitally. This will replace traditional paper-based methods, ensuring that transaction records are securely stored, easily accessible and well organized. While the input process still requires manual manipulation by the administrator, the module reduces the risk of losing or mismanaging records. It will also streamline the process by automatically classifying and storing transaction details for use by managers and members.

Transaction History Module

The transaction history module provides members with a secure and convenient interface to access personal records and services without relying on administrative assistance. Through this module, members can have real-time access to their transaction and transfer history, including saving deposit, loan deposit.

System Report Module

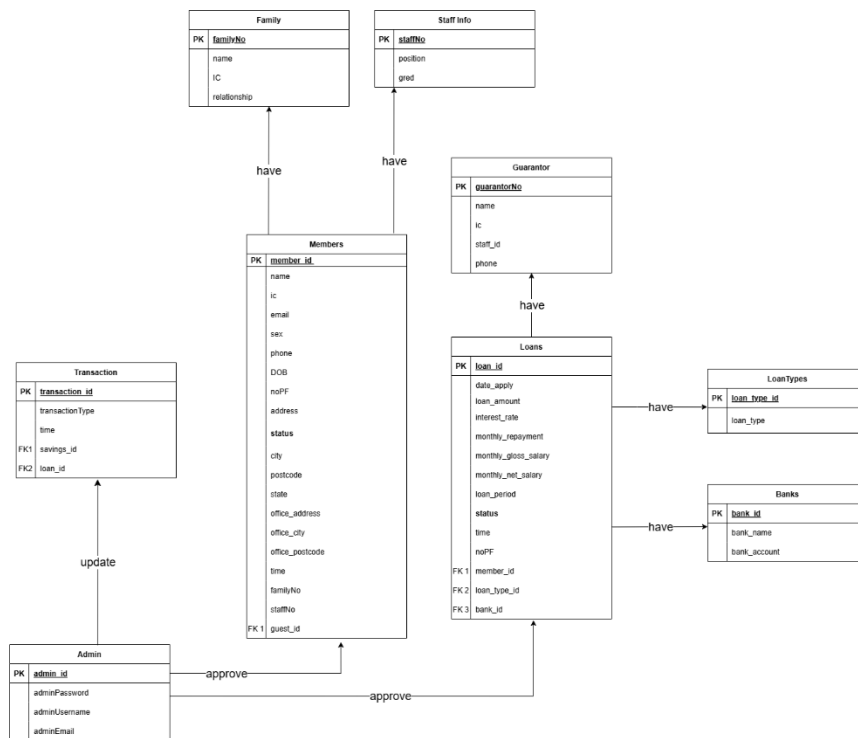
The system report module will introduce automation in the process of generating statistical reports, making it easier for the KADA team to monitor system performance and member activity. The main feature of this module is Visual Dashboards: This module will provide administrators with an interactive dashboard that allows them to see key metrics at a glance, such as the number of pending applications, the total number of transactions processed, and financial summaries. In addition, it includes an export feature that allows administrators to export system reports for use by analysts

Validation Module

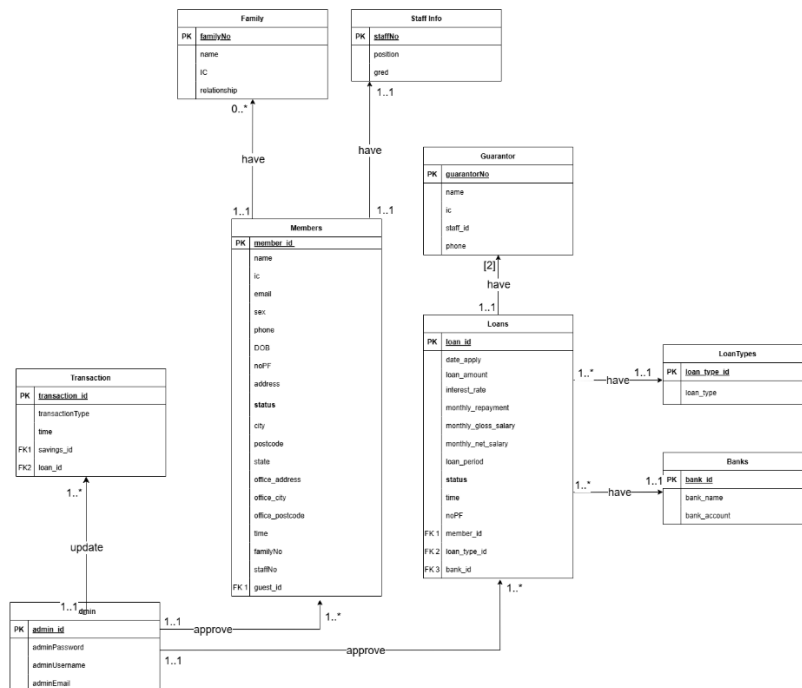
The proposed validation module will digitize and automate the process of validation applications, addressing the inefficiencies of paper-based approaches. The main functions of the module include automatic check. The module will automatically verify application details according to pre-set criteria, such as eligibility requirements, completeness of information and accuracy of data (e.g. telephone number must be filled in completely). The second one is status update. Once the application is verified, the system will automatically update the applicant's status on the member page status page to provide real-time updates for members.

SECTION B – DATABASE PLANNING AND DESIGN (40 MARKS)

2.1 Conceptual ERD



2.2 Logical ERD



2.3 Data Dictionary for Logical ERD

Entity Name	Description	Aliases	Occurrence
Member	General term describing all user logged in as member using KADA-eServe	Registered staff	Occurs when a guest registration application approved by admin
Admin	General term describing admin managing KADA-eServe	Clerk, Administrator, manager	Occurs when the system built
Transaction	General term describing all transaction record key-in by admin	Deposit amount, loan repayment	Occurs when admin key-in the amount of deposit or loan repayment for each member
Family	General term describing the family info of each member	Pewaris, Keluarga	Occurs when a guest registration application approved by admin
Staff Info	General term describing the staff position, grade info	KADA staff Info	Occurs when a guest registration application approved by admin
Guarantor	General term describing the guarantor of the member	Warrantor	Occurs when a guest registration application approved by admin
Loan	General term describing the loan info of member	Pembiayaan	Occurs when a member submits the loan application form to system
LoanTypes	General term describing the loan info of member	Loan type	Occurs when a member submits the loan application form to system
Banks	General term describing the bank account info of member	Bank account	Occurs when a member submits the loan application form to system

Identifying Relationship Types

Entity name	Multiplicity	Relationship	Entity Name	Multiplicity
Admin	1..1	approve	Members	1.. *
	1..1	approve	Loans	1.. *
Member	1..1	have	Staff Info	1..1
	1..1	have	Family	0.. *
	1..1	apply	Loans	1.. *
Loans	1..1	have	Guarantor	[2]
	1.. *	have	LoanTypes	1..1
	1.. *	have	Banks	1..1

Attributes Description

Entity Name	Attributes	Data Type & Length	Description	Nulls	Multi-valued	Example
Members	member_id (PK)	int	Uniquely identifies each member	No	No	5
	name	varchar(255)	Members' full name	No	No	Ahmad
	ic	int	Members' identification card number	No	No	981119-01-8732
	email	varchar(255)	Account email for receive notification	No	No	ahmad@gmail.com
	sex	varchar(255)	Members' gender	No	No	Lelaki
	phone	varchar(255)	Members' phone number	No	No	011-16596391
	DOB	date	Members' birth date	No	No	19/11/1998
	noPF	int	Members' personal profile	No	No	6
	address	text	Members' address	No	No	No 128 Taman U
	status	ENUM	Indicates the application state	No	No	Johor
	city	text	City where the member is located	No	No	Muar
	postcode	int	Postcode where the member is located	No	No	84000
	state	text	State where the member is located	No	No	Kelantan
	office_address	text	Office address of member	No	No	Peti Surat 127
	office_city	text	Office city of member	No	No	Bandar Kota Bharu
	office_postcode	int	Office postcode of member	No	No	15710
	time	date	The date of submission application	No	No	05/1/2025
	familyNo	int	Family record of members	Yes	Yes	2
	staffNo	int	Staff number of member	No	No	4672
Admin	admin_id (PK)	int	Uniquely identifies each admin	No	No	1
	adminPassword	varchar(255)	Account password for admin login	No	No	\$gf@142@
	adminUsername	varchar(255)	Admins' full name	No	No	Ahmad
	adminEmail	varchar(255)	Account email for admin login	No	No	admin@gmail.com
Transaction	Transaction_id (PK)	int	Uniquely identified transaction record	No	No	6
	transactionType	ENUM	Type of transaction made	No	Yes	Loan repay , Deposit
	time	date	Date of transaction made	No	No	19/1/2025

	Savings_id (FK)	int	Indicated to the specific savings account info	No	No	7
	Loan_id (FK)	int	Indicated to the specific loan application for repayment	No	No	LOAN-1737381950
Family	FamilyNo (PK)	int	Uniquely identified family record of member	No	No	4
	name	varchar(255)	Members' family name	No	No	Ismail
	IC	varchar(255)	Members' family ic	No	No	931118-01-1236
	relationship	varchar(255)	Relationship between the family and member	No	No	Adik lelaki
Staff Info	staffNo	int	Uniquely identified each staff	No	No	9824
	position	varchar(255)	Staffs' position in KADA	No	No	perkhidmatan pelanggan
	graded	varchar(255)	Staffs' grade in KADA	No	No	M40
Guarantor	GuarantorNo (PK)	int	Uniquely identified guarantor of member	No	No	7
	name	varchar(255)	Guarantors' name	No	No	Siti
	ic	varchar(255)	Guarantors' IC	No	No	991231-01-9823
	staff_id	int	Guarantors' staff ID in KADA	No	No	7
	phone	varchar(255)	Guarantors' phone number	No	No	019-7113981
Loans	Loan_id (PK)	int	Uniquely identified each application record	No	No	6
	date_apply	date	Submission date of application	No	No	20/1/2025
	loan_amount	float	Amount of loan to apply	No	No	2000.50
	interest_rate	float	Interest rate for loan	No	No	0.05
	monthly_repayment	float	Monthly repayment of member	No	No	200
	monthly_gross_salary	float	Monthly gross salary of member	No	No	4000
	monthly_net_salary	float	Monthly net salary of member	No	No	3500
	loan_period	date	Period of loan apply	No	No	18/2/2028
	status	ENUM	Status of the application	No	No	approved
	time	date	Submission date of the application	No	No	19/1/2025
	noPF	int	Indicated to the personal profile of member	No	No	4
	member_id (FK 1)	int	Indicated to member ID	No	No	8
	loan_type_id (FK 2)	int	indicated to type of loan member apply	No	No	4
	bank_id (FK 3)	int	Indicated to bank record of member	No	No	6

LoanTypes	loan_type_id (PK)	int	Uniquely identified the type of loan	No	No	5
	loan_type	varchar(255)	Type of loan to be applied	No	No	Simpanan Tetap

2.4 Relational Database Schema

The Relational Database Schema represents the tables derived from the Logical ERD, showing the primary keys, foreign keys and relationships. Below is the relational database schema for this Administrative Module:

1. Member (member_id, name, ic, email, sex, phone, DOB, noPF, address, status, city, postcode, state, office_address, office_city, office_postcode, time, familyNo, staffNo, guest_id)
Primary Key: member_id
Foreign Key: guest_id references Guest(guest_id), familyNo references Family(familyNo), staffNo references StaffInfo(staffNo)
2. Admin (admin_id, adminPassword, adminUsername, adminEmail)
Primary Key: admin_id
3. Family (familyNo, name, IC, relationship)
Primary Key: familyNo
4. Staff Info (staffNo, position, gred)
Primary Key: staffNo
5. Guarantors (guarantor_no, name, ic, phone, staffID)
Primary Key: guarantor_id
6. Loans (loan_id, member_id, loan_type_id, bank_id, date_apply, loan_amount, interest_rate, monthly_repayment, monthly_gross_salary, monthly_net_salary, loan_period, status, time ,noPF)
Primary Key: loan_id
Foreign Keys: member_id references Member (member_id), loan_type_id references Loan_Types(loan_type_id)
7. Loan_Types (loan_type_id, loan_type)
Primary Key: loan_type_id
8. Transactions (Transaction_id, member_id, Transaction_type, savings_type, loan_id, amount, time)
Primary Key: Transaction_id
Foreign Keys: member_id references Member_Register(id), loan_id references Loans(loan_id)

2.5 Normalization

Member Table

1NF

Member (member_id, name, ic, email, sex, phone, DOB, noPF, address, status,city, postcode, state, office_address, office_city, office_postcode, time, staffNo,guest_id, family_name, family_IC, relationship, position, gred)

Primary Key: member_id

2NF

Member (member_id, name, ic, email, sex, phone, DOB, noPF, address, status,city, postcode, state, office_address, office_city, office_postcode, time, staffNo,guest_id,familyNo, position, gred)

Primary Key: member_id

Foreign Key: familyNo references Family(familyNo)

Family (familyNo, name, IC, relationship)

Primary Key: familyNo

3NF

Member (member_id, name, ic, email, sex, phone, DOB, noPF, address, status,city, postcode, state, office_address, office_city, office_postcode, time, guest_id, staffNo, familyNo)

Primary Key: member_id

Foreign Key: guest_id references Guest(guest_id), familyNo references Family(familyNo), staffNo references StaffInfo(staffNo)

Family (familyNo, name, IC, relationship)

Primary Key: familyNo

StaffInfo (staffNo, position, gred)

Primary Key: staffNo

Admin Table

1NF

Admin (admin_id, adminPassword, adminUsername, adminEmail)

Primary Key: admin_id

2NF

Since there are no partial dependencies in the relation, all derived relations are already in second normal form (2NF). Therefore, all derived dependencies will remain the same, as in the first normal form (1NF).

3NF

Since there are no partial dependencies in the relation, all derived relations are already in third normal form (3NF). Therefore, all derived dependencies will remain the same, as in the first normal form (1NF).

Transaction table

1NF

Transactions (Transaction_id, name, ic, email, sex, phone, DOB, noPF, address, status,city, postcode, state, office_address, office_city, office_postcode, time, guest_id, family_name, family_IC, relationship, position, gred, Transaction_type, loan_id, member_id, loan_type_id, bank_id, date_apply, loan_amount, interest_rate, monthly_repayment, monthly_gross_salary, monthly_net_salary, loan_period, status, time, noP , amount, time, loan_type, guarantor_name, guarantor_ic, guarantor_phone,guarantor_staffNo)

Primary Key: Transaction_id

2NF

Transactions (Transaction_id, member_id, Transaction_type, loan_id, amount, time)

Primary Key: Transaction_id

Foreign Keys:member_id references Member (member_id),loan_id references Loans(loan_id)

Member (member_id, name, ic, email, sex, phone, DOB, noPF, address, status,city, postcode, state, office_address, office_city, office_postcode, time, staffNo,guest_id, family_name, family_IC, relationship, position, gred)

Primary Key: member_id

Loans (loan_id, member_id, loan_type, bank_id, date_apply, loan_amount, interest_rate, monthly_repayment, monthly_gross_salary, monthly_net_salary, loan_period, status, time ,noPF, guarantor_name, guarantor_ic, guarantor_phone,guarantor_staffNo)

Primary Key: loan_id

Foreign Key: member_id references Member (member_id)

3NF

Transactions (Transaction_id, member_id, Transaction_type, loan_id, amount, time)

Primary Key: Transaction_id

Foreign Keys:member_id references Member (member_id),loan_id references Loans(loan_id)

Member (member_id, name, ic, email, sex, phone, DOB, noPF, address, status,city, postcode, state, office_address, office_city, office_postcode, time, guest_id, staffNo, familyNo)

Primary Key: member_id

Foreign Key: guest_id references Guest(guest_id), familyNo references Family(familyNo), staffNo references StaffInfo(staffNo)

Family (familyNo, name, IC, relationship)

Primary Key: familyNo

StaffInfo (staffNo, position, gred)

Primary Key: staffNo

Loans (loan_id, member_id, loan_type_id, bank_id, date_apply, loan_amount, interest_rate, monthly_repayment, monthly_gross_salary, monthly_net_salary, loan_period, status, time ,noPF, guarantor_no)

Primary Key: loan_id

Foreign Keys:member_id references Member (member_id),loan_type_id references
Loan_Types(loan_type_id),guarantor_no references Guarantors(guarantor_no)

Loan_Types (loan_type_id, loan_type)

Primary Key: loan_type_id

Guarantors (guarantor_no, name, ic, phone, staffID)

Primary Key: guarantor_id